

Faculty of Information Technology

Fall 2025/2026

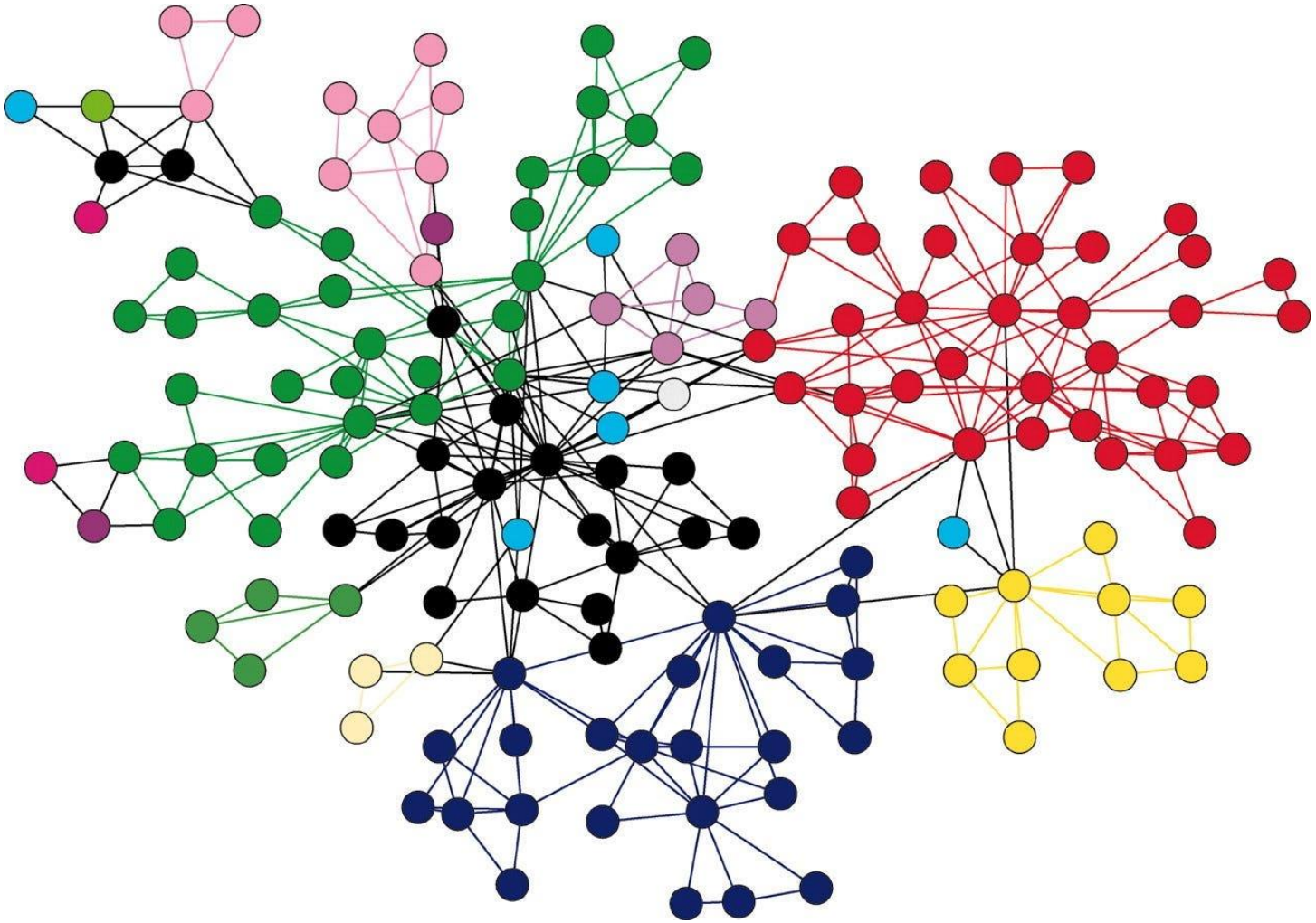
DATA STRUCTURES

CS.212 / CS-212

Lec (7)

- By Dr. Alaa Zaghloul

Graph



Introduction:

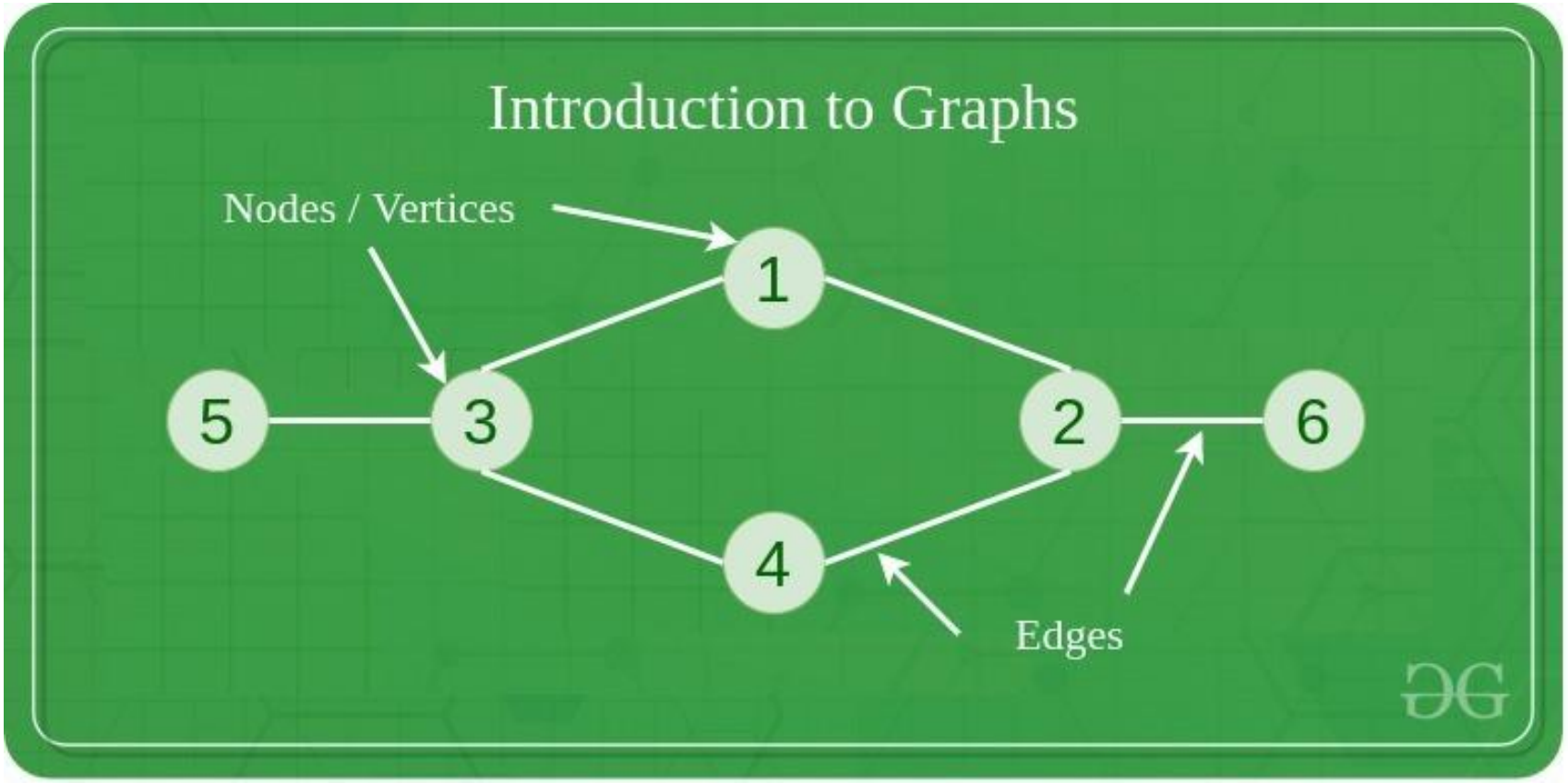
A Graph is a non-linear data structure consisting of vertices and edges.

The vertices are sometimes also referred to as nodes and the edges are lines or arcs that connect any two nodes in the graph.

More formally a Graph is composed of a set of vertices(**V**) and a set of edges(**E**).

The graph is denoted by **G(V, E).**

Components of a Graph



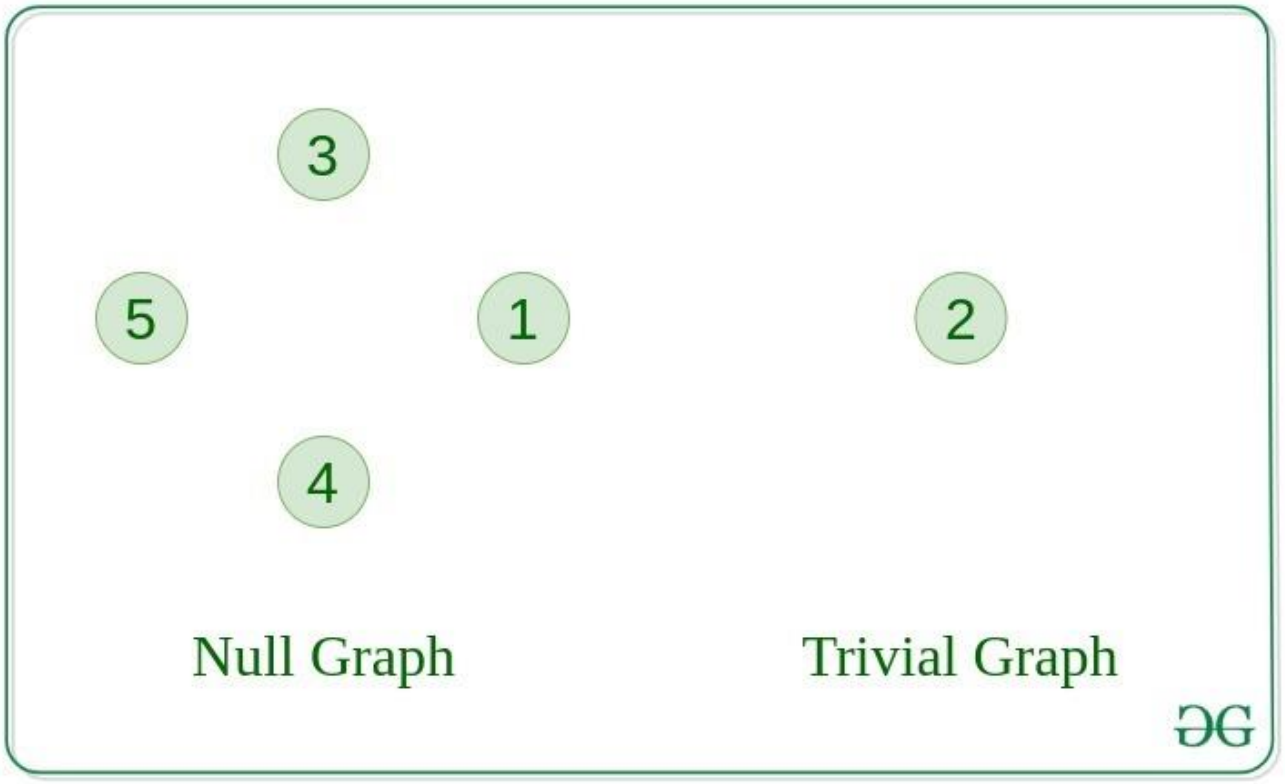
Types Of Graph

1. Null Graph

A graph is known as a null graph if there are no edges in the graph.

2. Trivial Graph

Graph having only a single vertex, it is also the smallest graph possible.



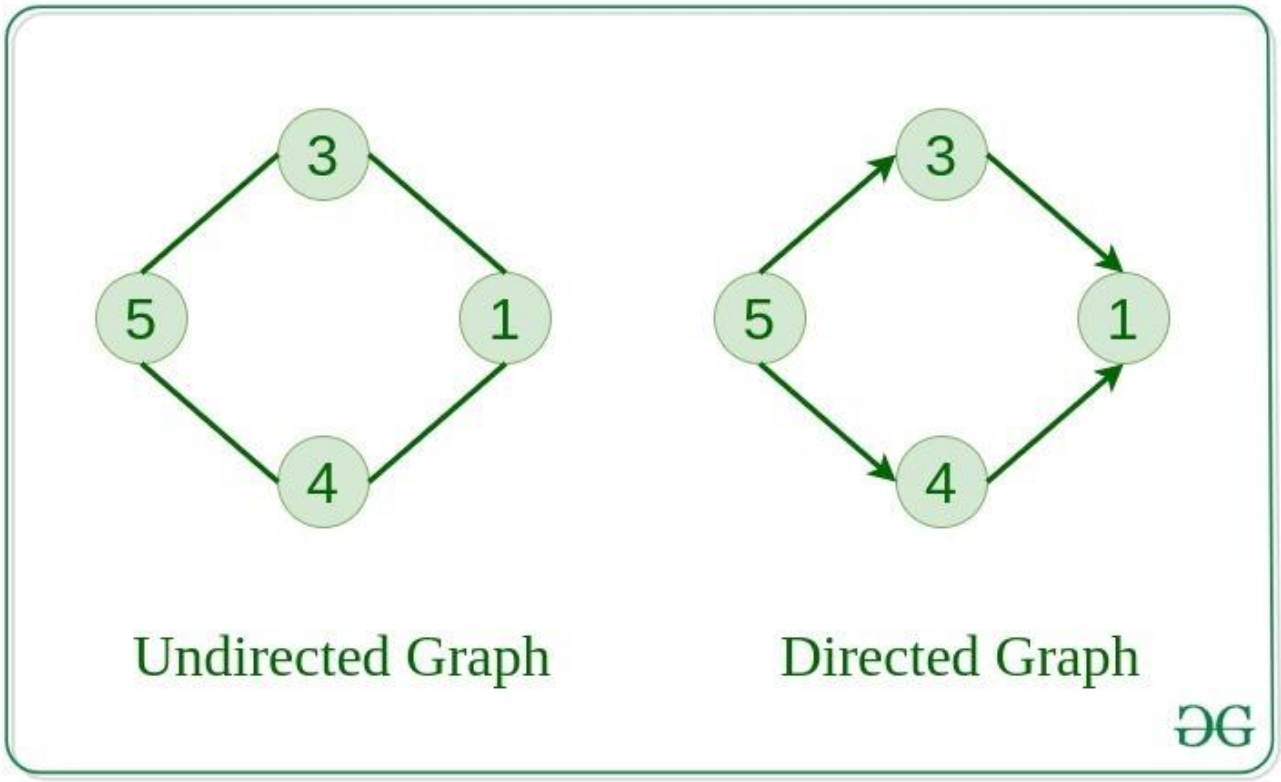
Types Of Graph

3. Undirected Graph

A graph in which edges do not have any direction. That is the nodes are unordered pairs in the definition of every edge.

4. Directed Graph

A graph in which edge has direction. That is the nodes are ordered pairs in the definition of every edge.



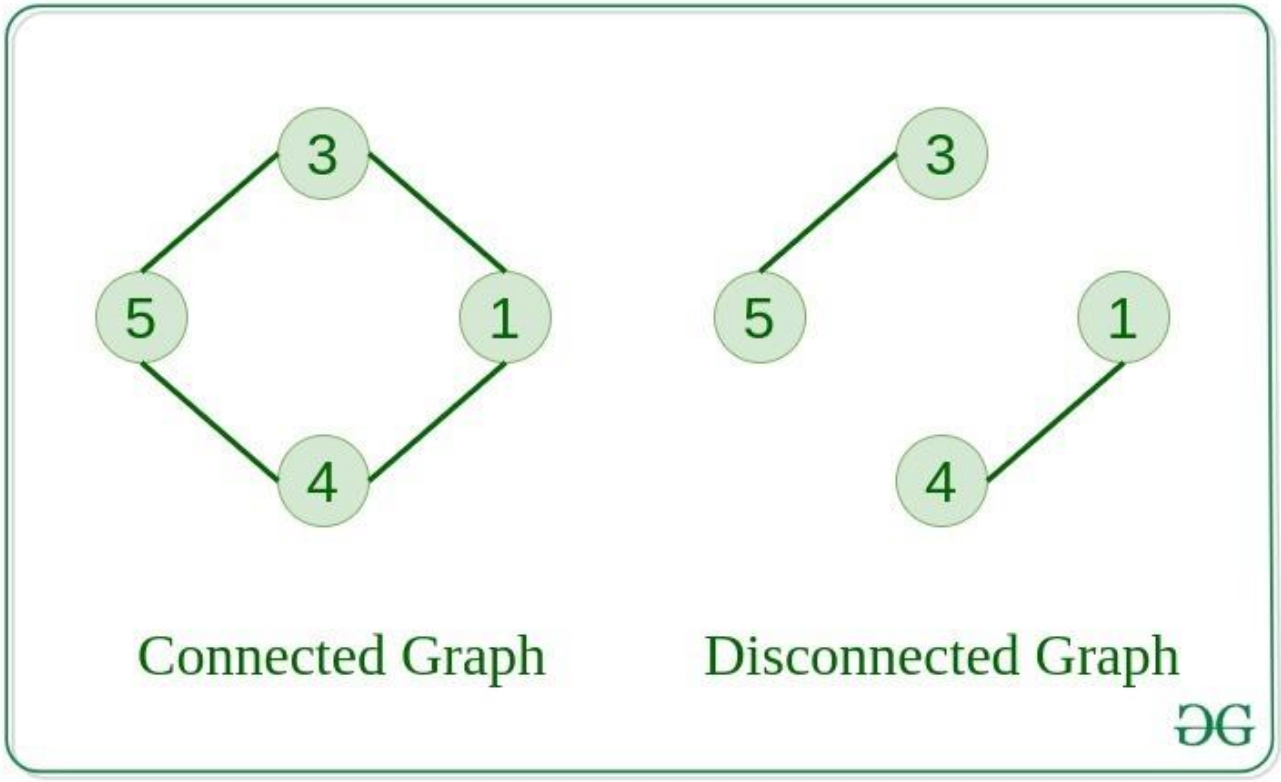
Types Of Graph

5. Connected Graph

The graph in which from one node we can visit any other node in the graph is known as a connected graph.

6. Disconnected Graph

The graph in which at least one node is not reachable from a node is known as a disconnected graph.



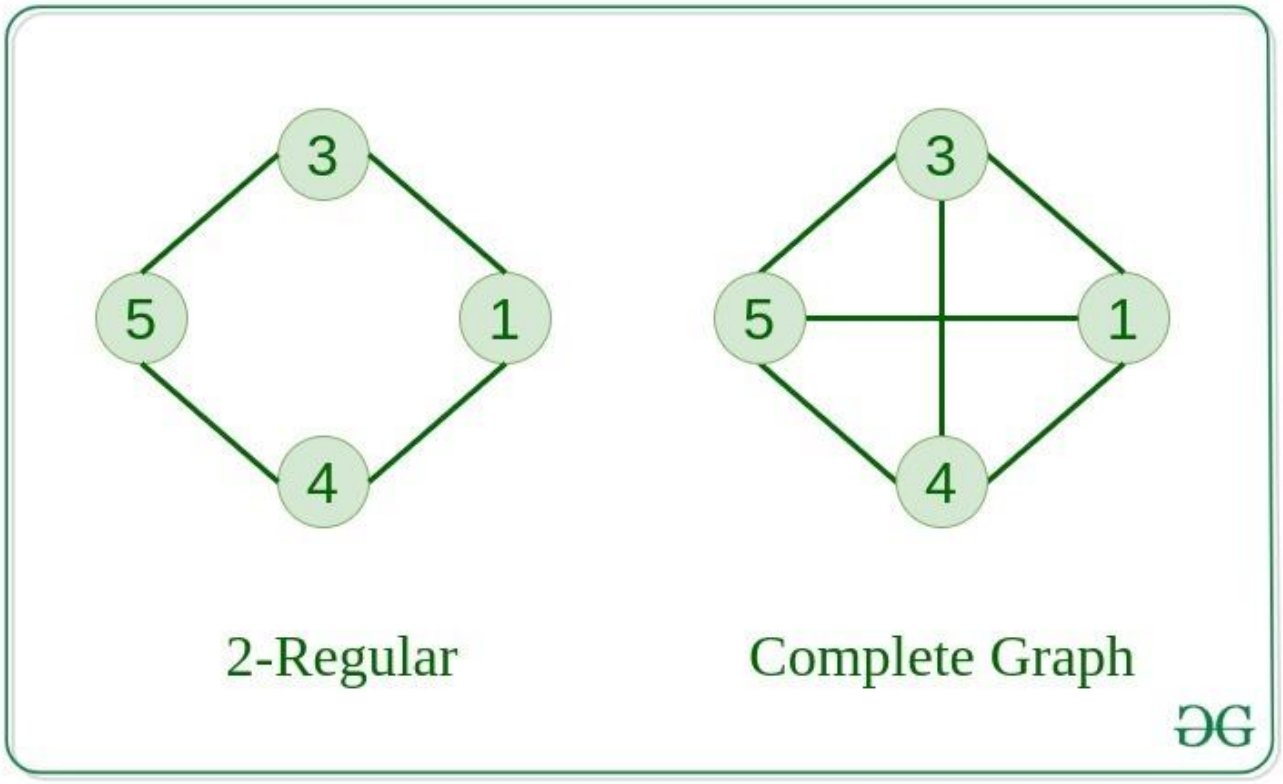
Types Of Graph

7. Regular Graph

The graph in which the degree of every vertex is equal to K is called K regular graph.

8. Complete Graph

The graph in which from each node there is an edge to each other node.



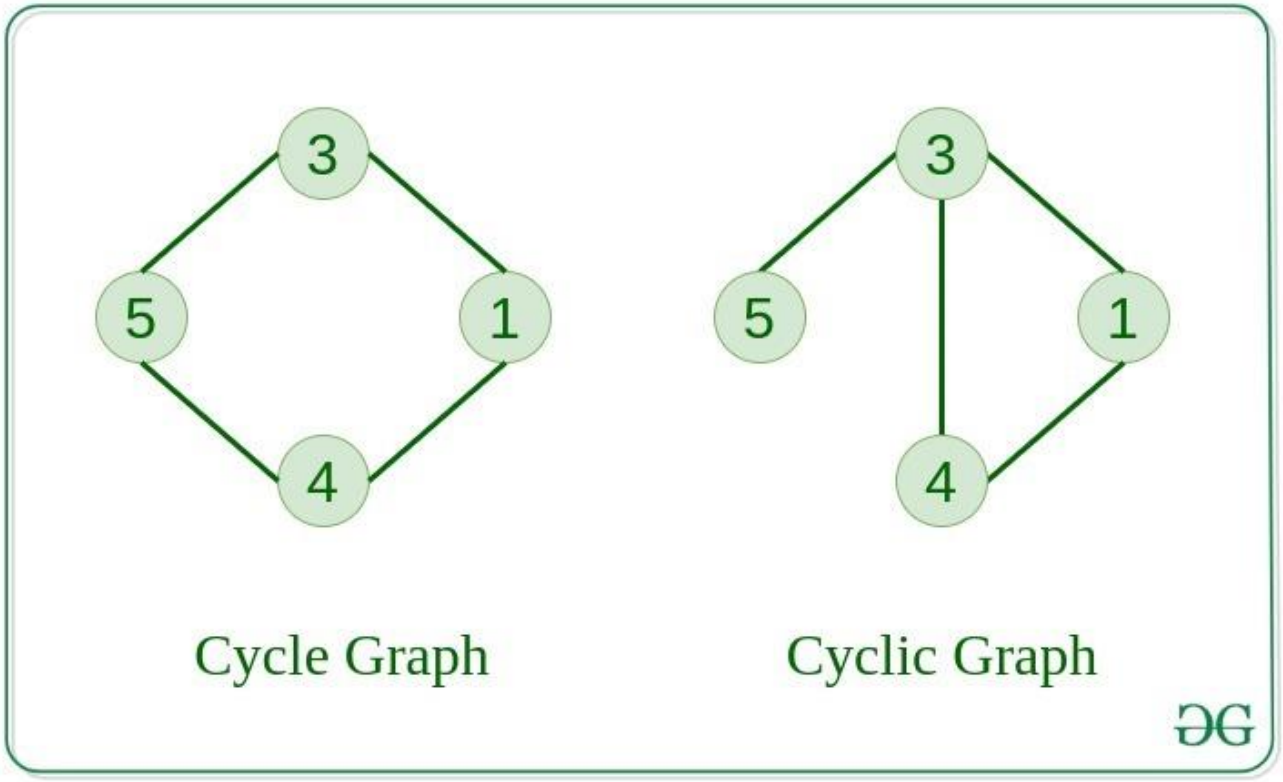
Types Of Graph

9. Cycle Graph

The graph in which the graph is a cycle in itself, the degree of each vertex is 2.

10. Cyclic Graph

A graph containing at least one cycle is known as a Cyclic graph.



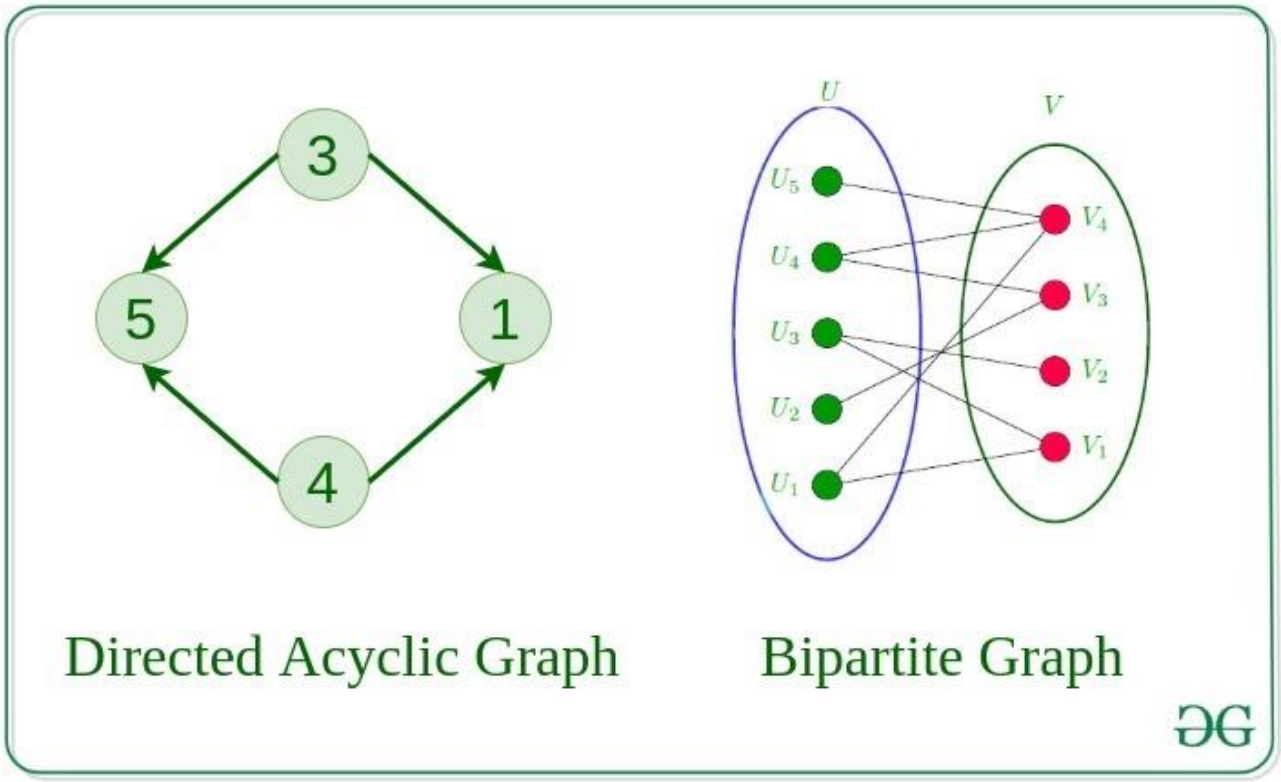
Types Of Graph

11. Directed Acyclic Graph

A Directed Graph that does not contain any cycle.

12. Bipartite Graph

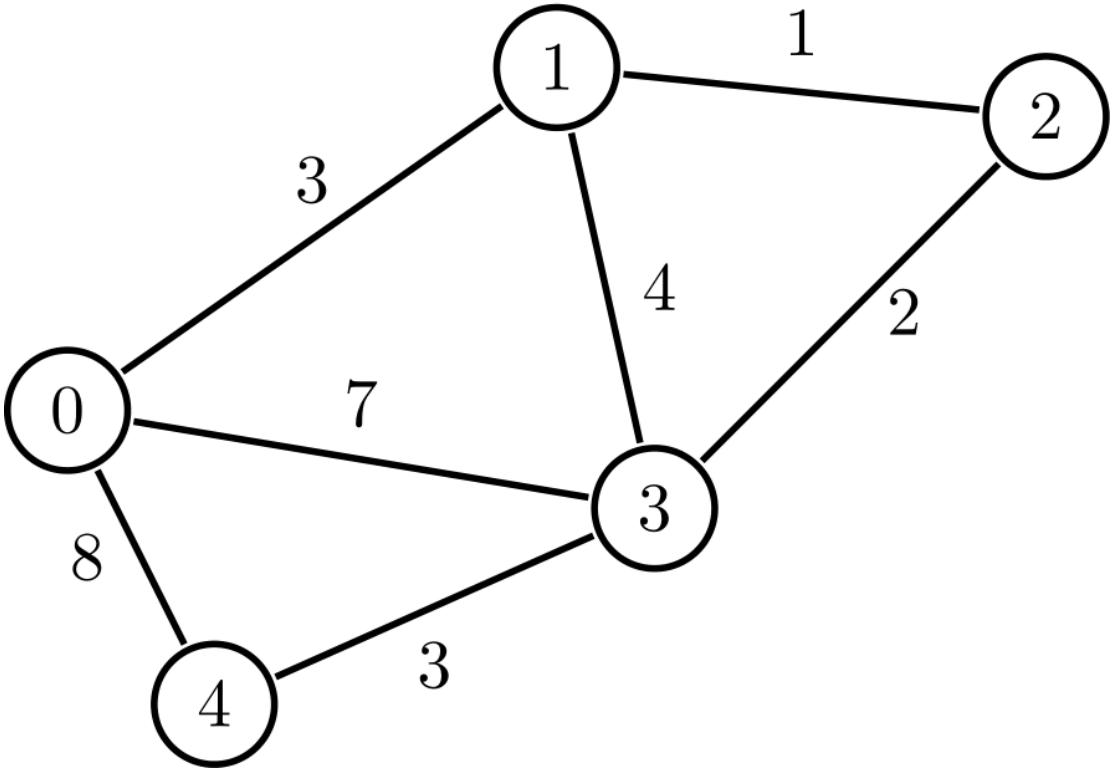
A graph in which vertex can be divided into two sets such that vertex in each set does not contain any edge between them.



Types Of Graph

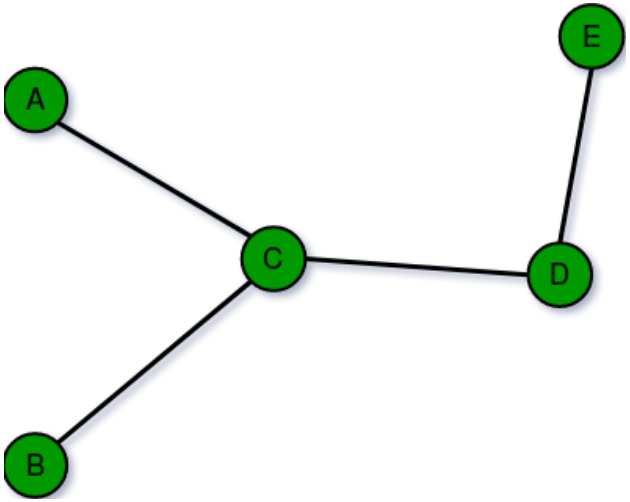
13. Weighted Graph

- A graph in which the edges are already specified with suitable weight is known as a weighted graph.
- Weighted graphs can be further classified as directed weighted graphs and undirected weighted graphs.

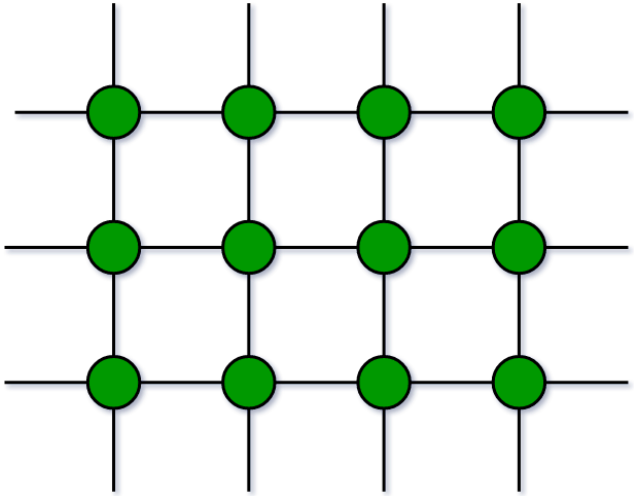


Types Of Graph

Finite Graphs

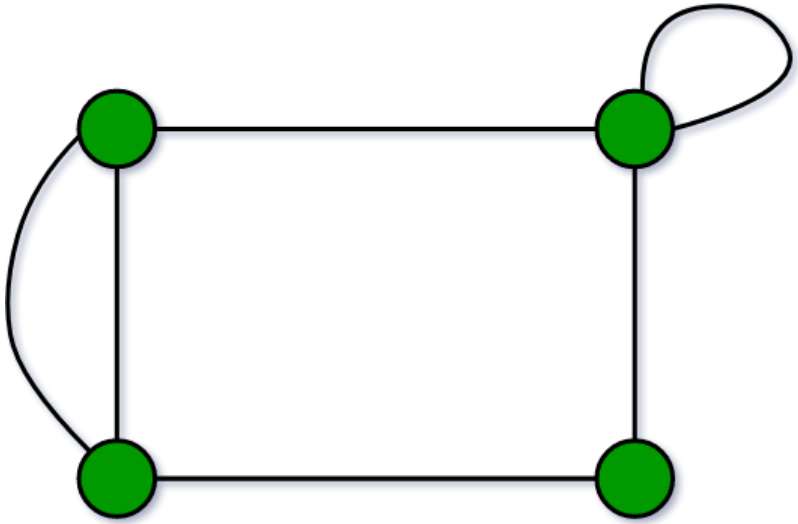


Infinite Graph

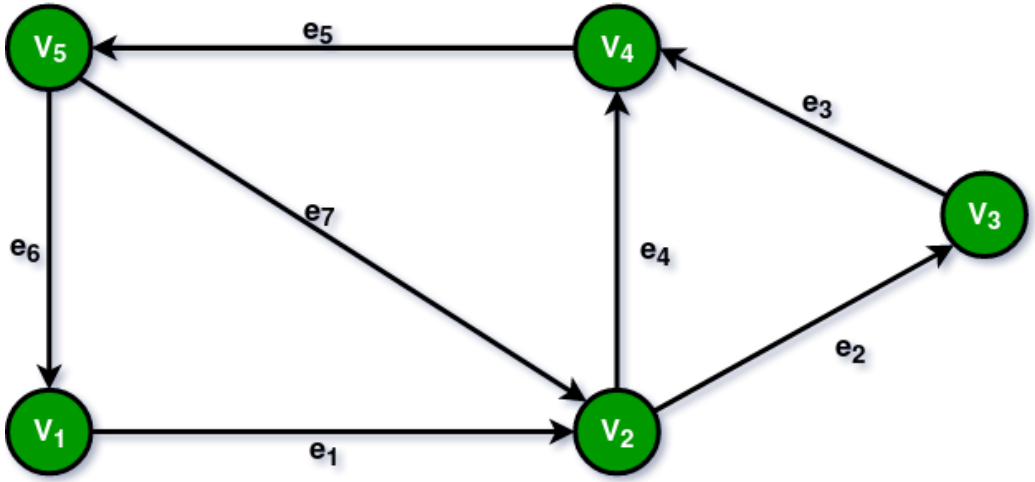


Types Of Graph

Pseudo Graph



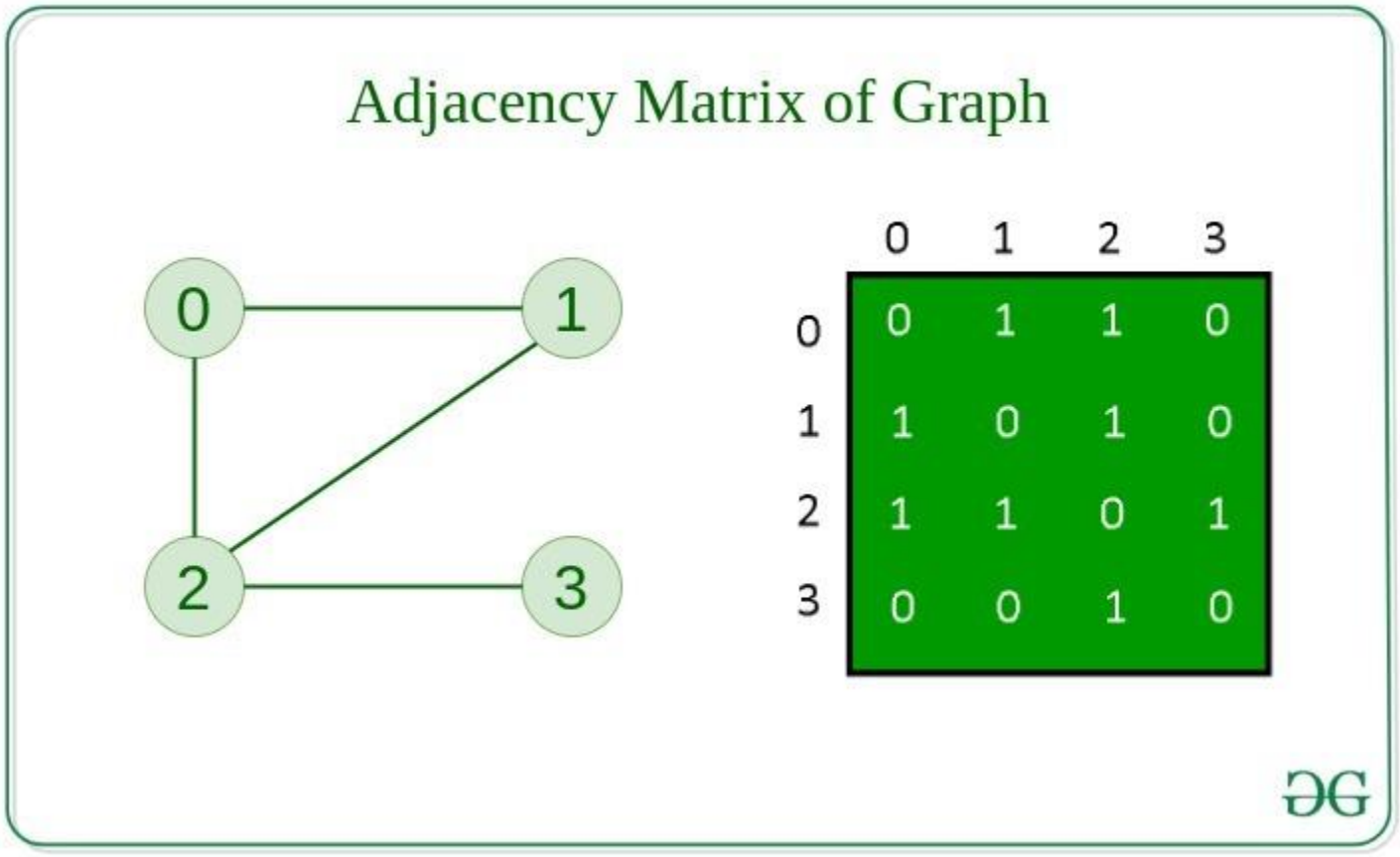
Labeled Graph



Representation Of Graph

There are two ways to store a graph:

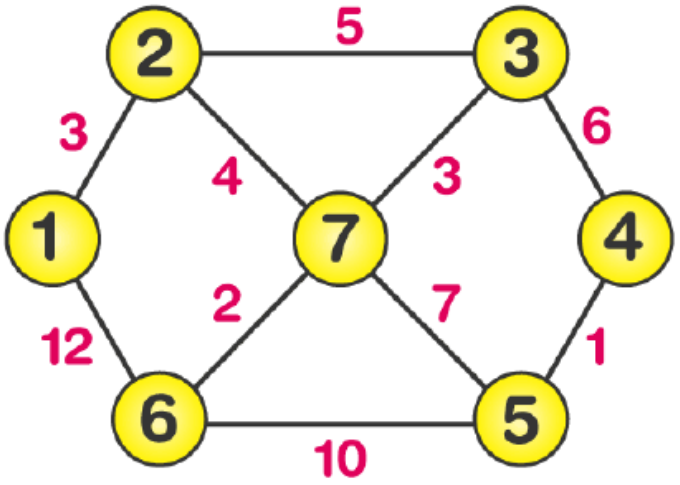
- Adjacency Matrix
- Adjacency List



Representation Of Graph

There are two ways to store a graph:

- Adjacency Matrix
- Adjacency List



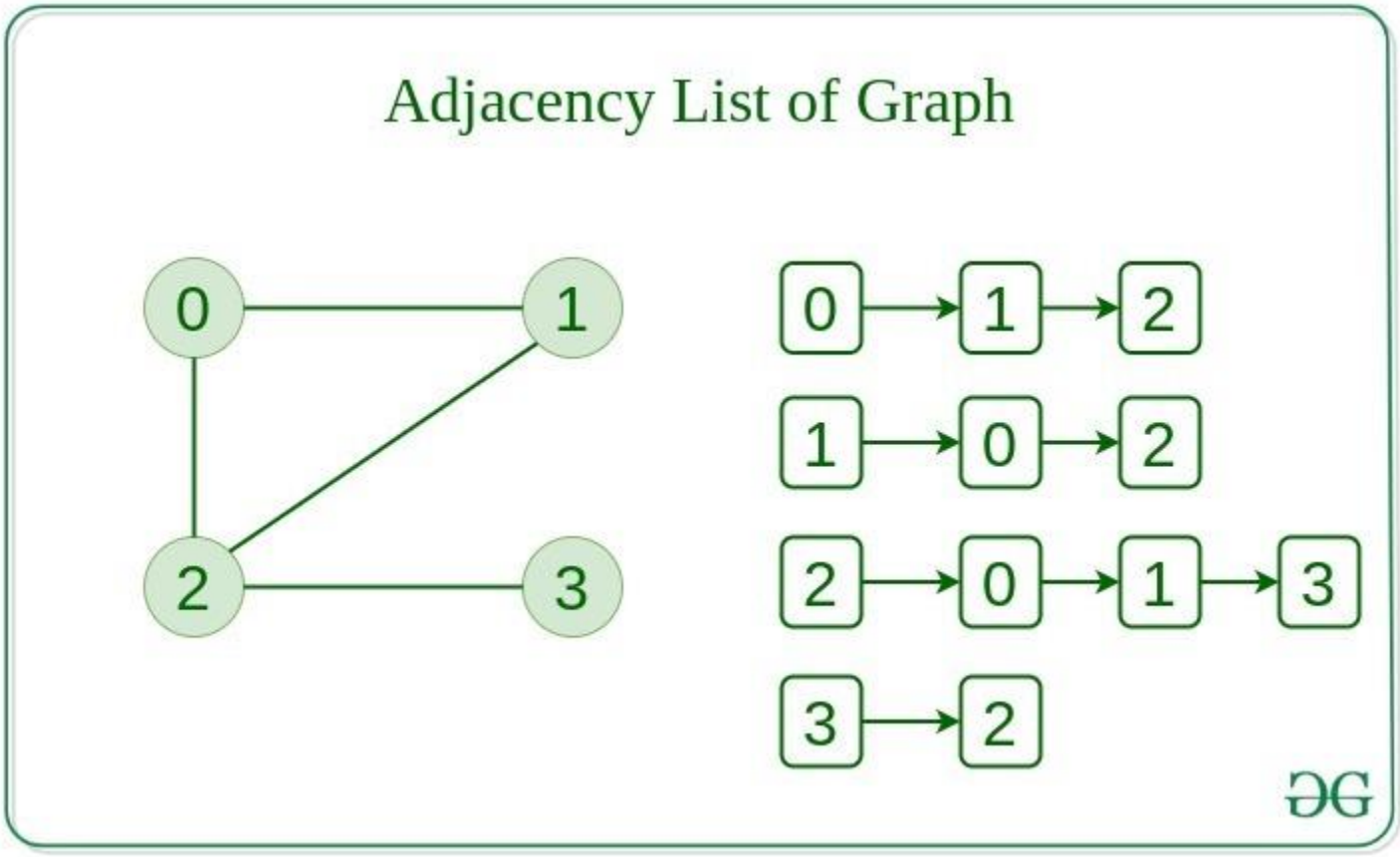
© Byjus.com

$$A = \begin{bmatrix} 0 & 3 & 0 & 0 & 0 & 12 & 0 \\ 3 & 0 & 5 & 0 & 0 & 0 & 4 \\ 0 & 5 & 0 & 6 & 0 & 0 & 3 \\ 0 & 0 & 6 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 10 & 7 \\ 12 & 0 & 0 & 0 & 10 & 0 & 2 \\ 0 & 4 & 3 & 0 & 7 & 2 & 0 \end{bmatrix}$$

Representation Of Graph

There are two ways to store a graph:

- Adjacency Matrix
- Adjacency List

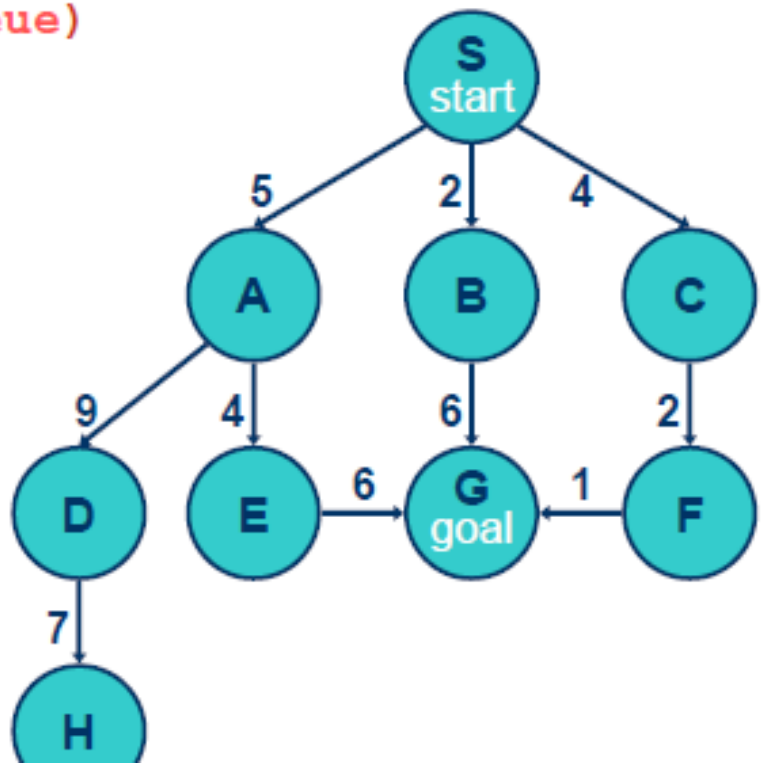


Breadth-First Search

```
generalSearch(problem, queue)
```

of nodes tested: 0, expanded: 0

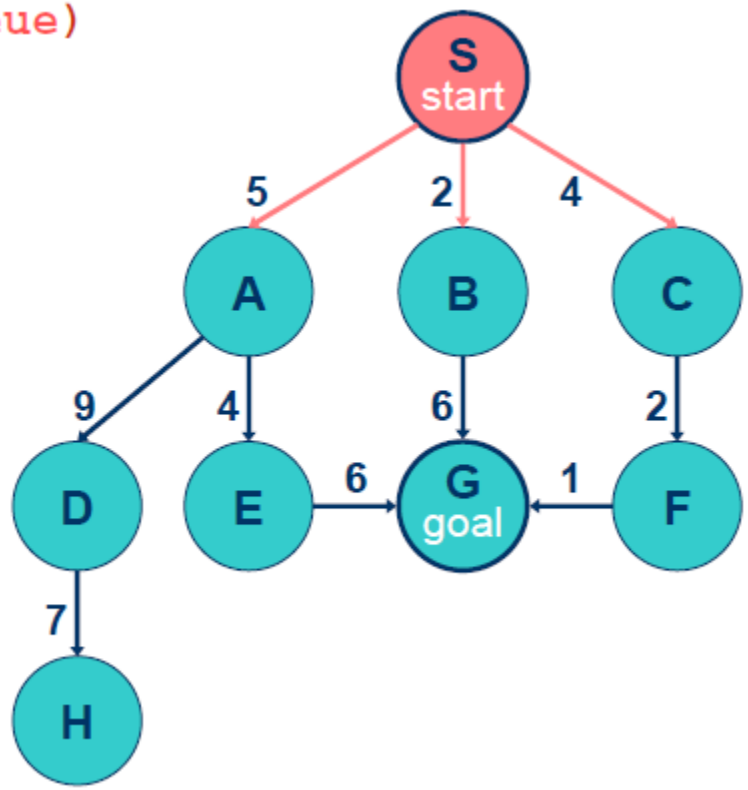
expnd. node	nodes list
	{S}



`generalSearch(problem, queue)`

of nodes tested: 1, expanded: 1

expnd. node	nodes list
	{S}
S not goal	{A,B,C}

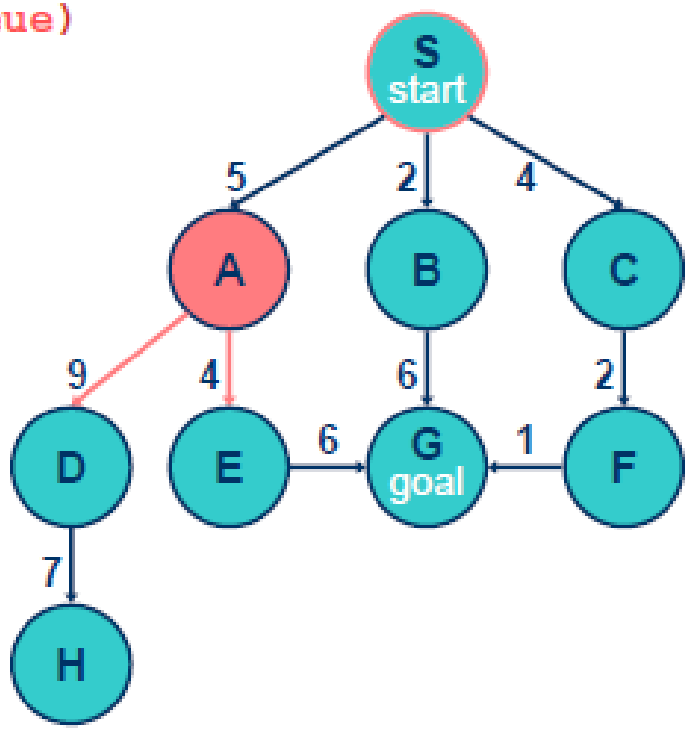


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`generalSearch(problem, queue)`

of nodes tested: 2, expanded: 2

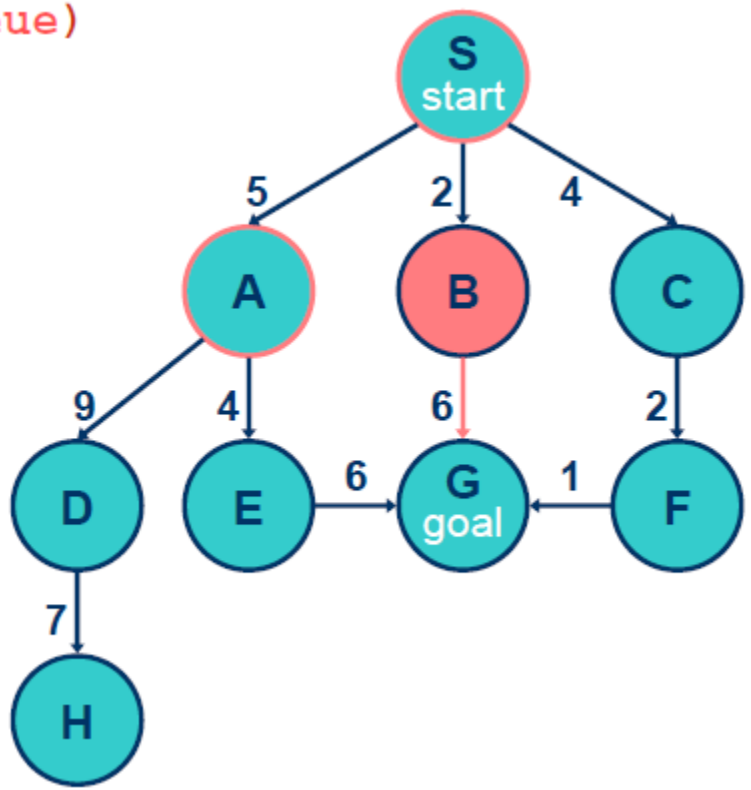
expnd. node	nodes list
	{S}
S	{A,B,C}
A not goal	{B,C,D,E}



`generalSearch(problem, queue)`

of nodes tested: 3, expanded: 3

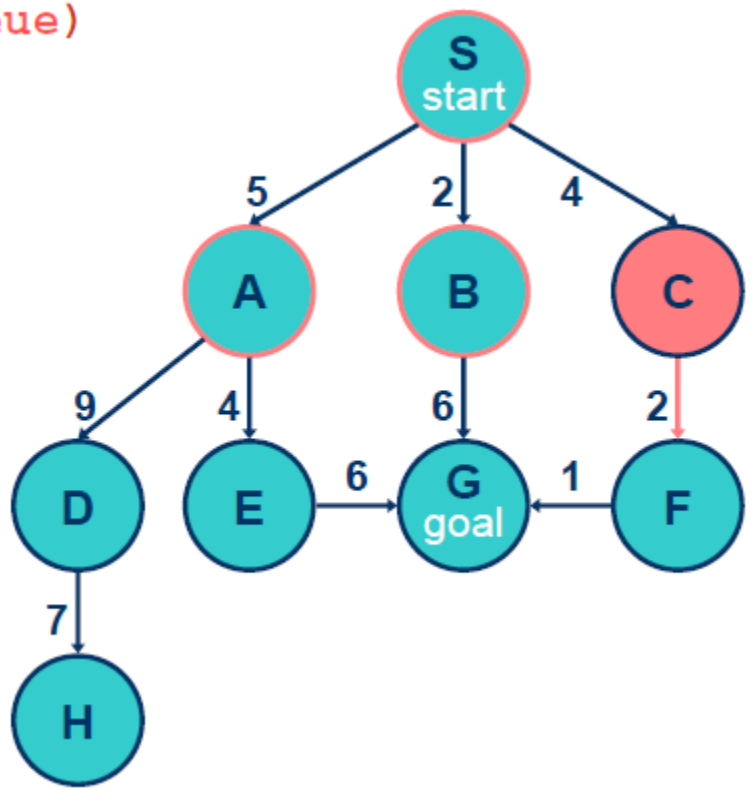
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B not goal	{C,D,E,G}



`generalSearch(problem, queue)`

of nodes tested: 4, expanded: 4

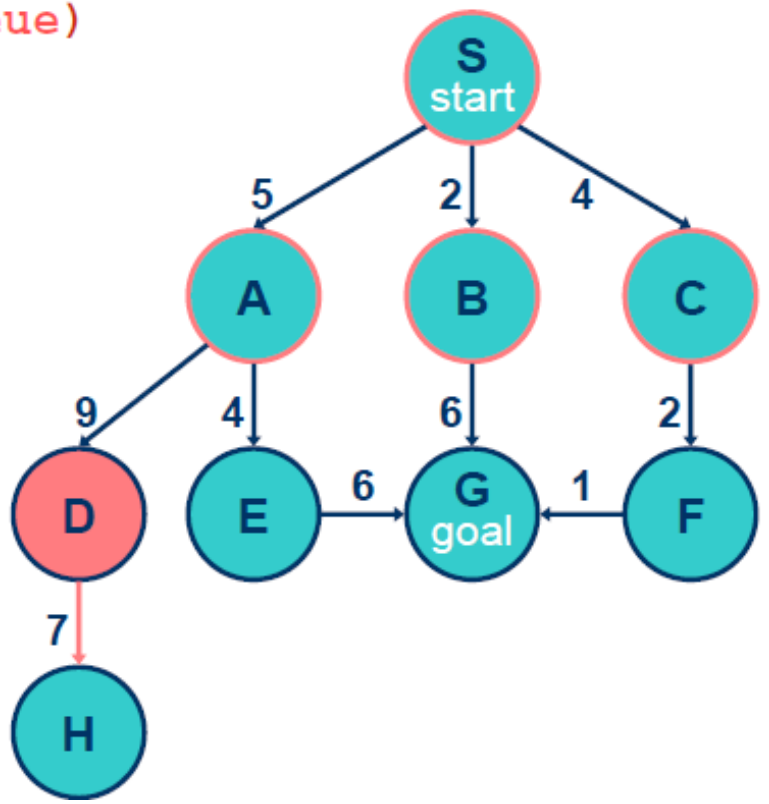
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B	{C,D,E,G}
C not goal	{D,E,G,F}



`generalSearch(problem, queue)`

of nodes tested: 5, expanded: 5

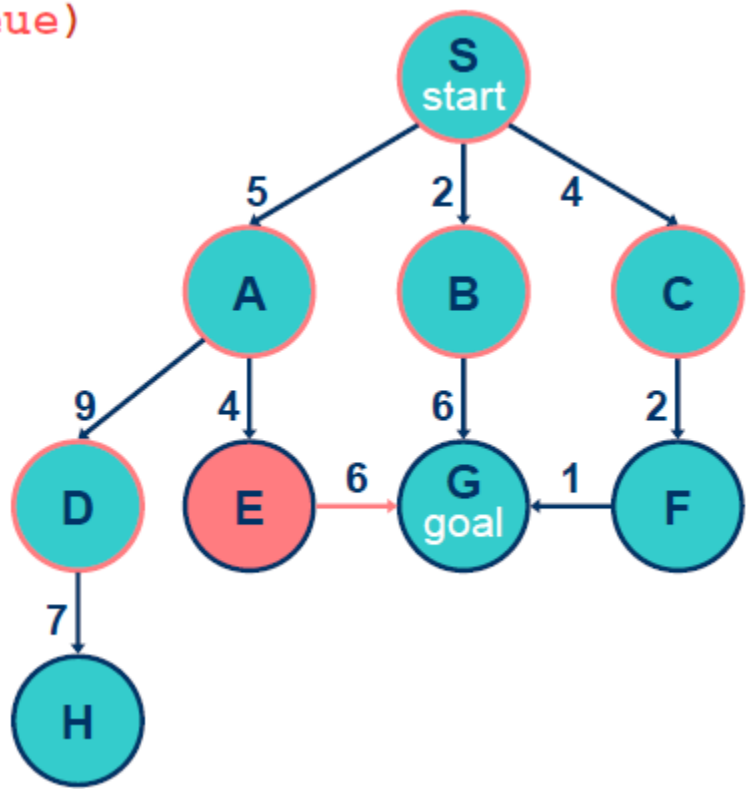
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B	{C,D,E,G}
C	{D,E,G,F}
D not goal	{E,G,F,H}



`generalSearch(problem, queue)`

of nodes tested: 6, expanded: 6

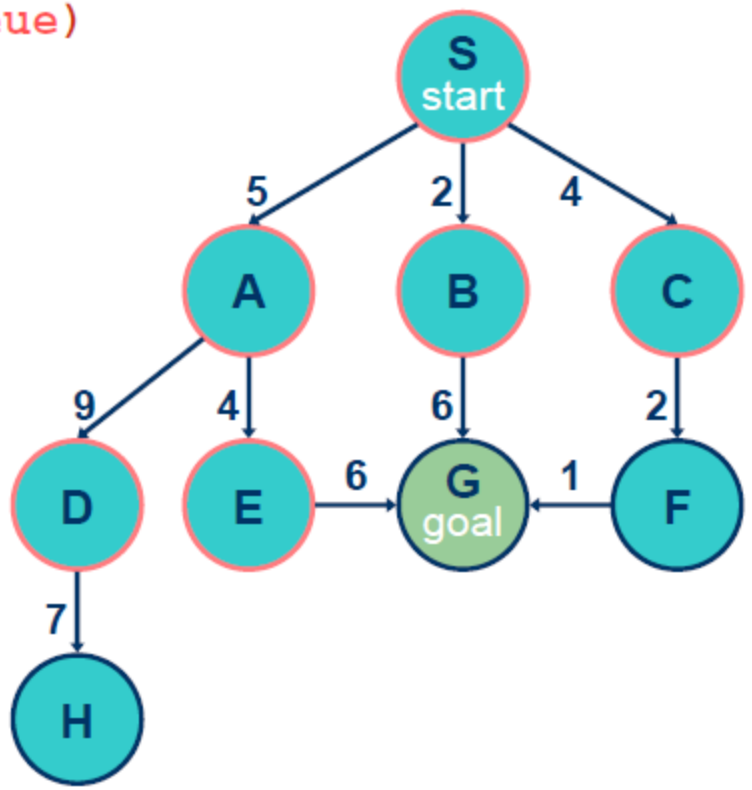
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B	{C,D,E,G}
C	{D,E,G,F}
D	{E,G,F,H}
E not goal	{G,F,H,G}



`generalSearch(problem, queue)`

of nodes tested: 7, expanded: 6

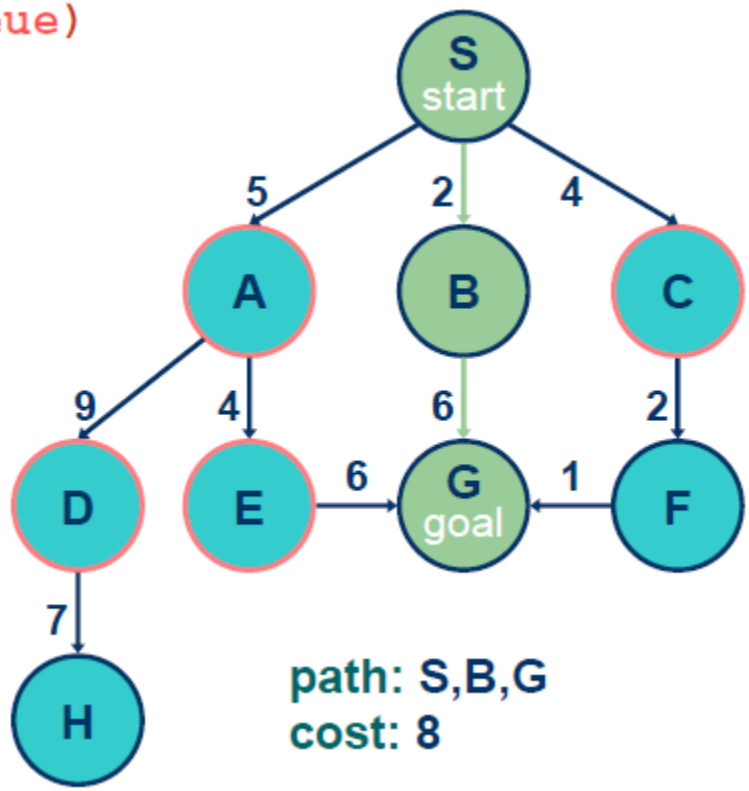
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B	{C,D,E,G}
C	{D,E,G,F}
D	{E,G,F,H}
E	{G,F,H,G}
G goal	{F,H,G} no expand



`generalSearch(problem, queue)`

of nodes tested: 7, expanded: 6

expnd. node	nodes list
	{S}
S	{A,B,C}
A	{B,C,D,E}
B	{C,D,E,G}
C	{D,E,G,F}
D	{E,G,F,H}
E	{G,F,H,G}
G	{F,H,G}

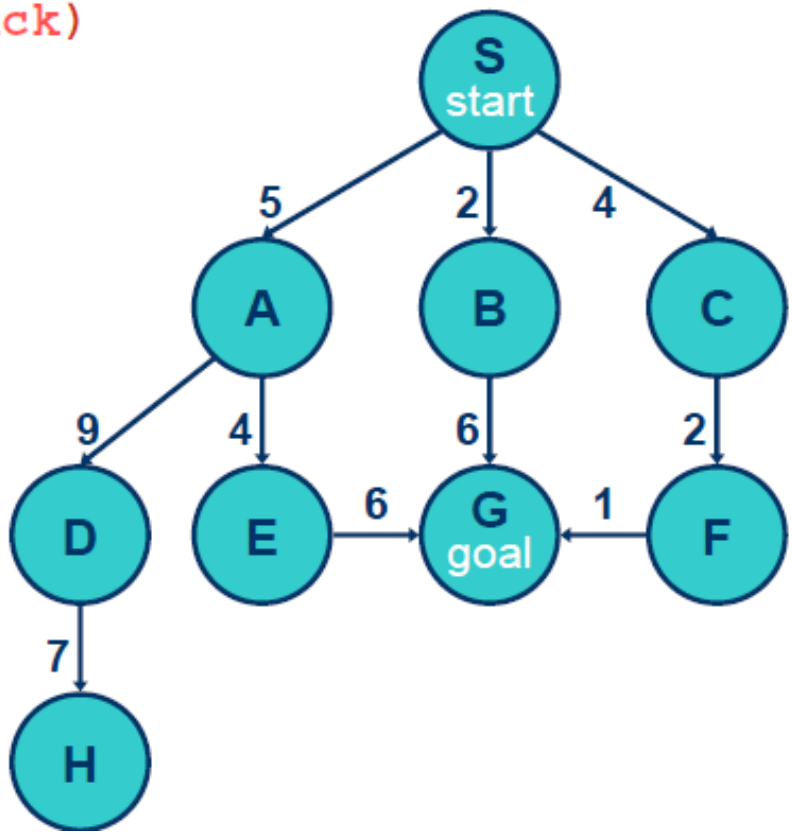


Depth-first

`generalSearch(problem, stack)`

of nodes tested: 0, expanded: 0

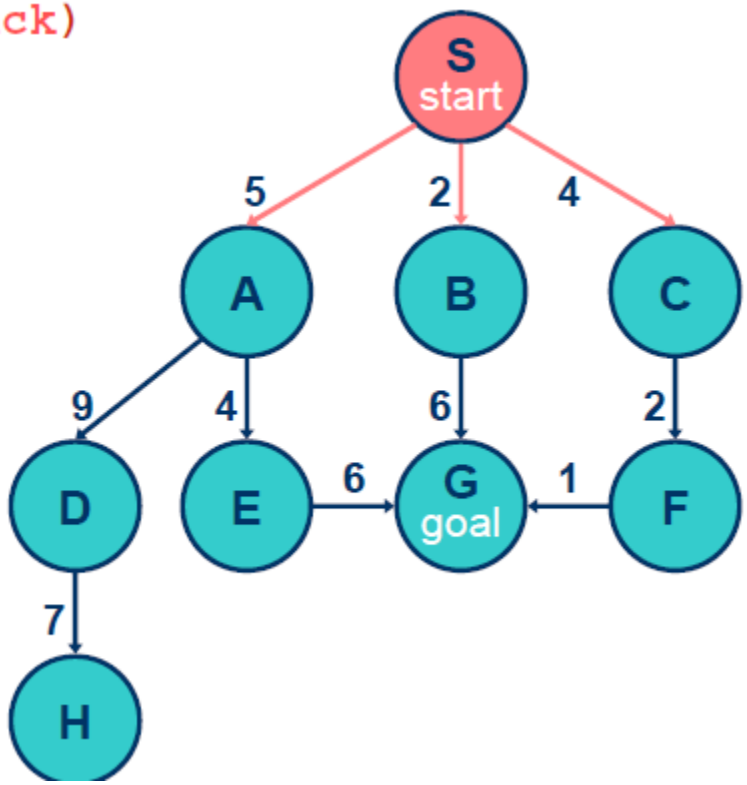
expnd. node	nodes list
	{S}



`generalSearch(problem, stack)`

of nodes tested: 1, expanded: 1

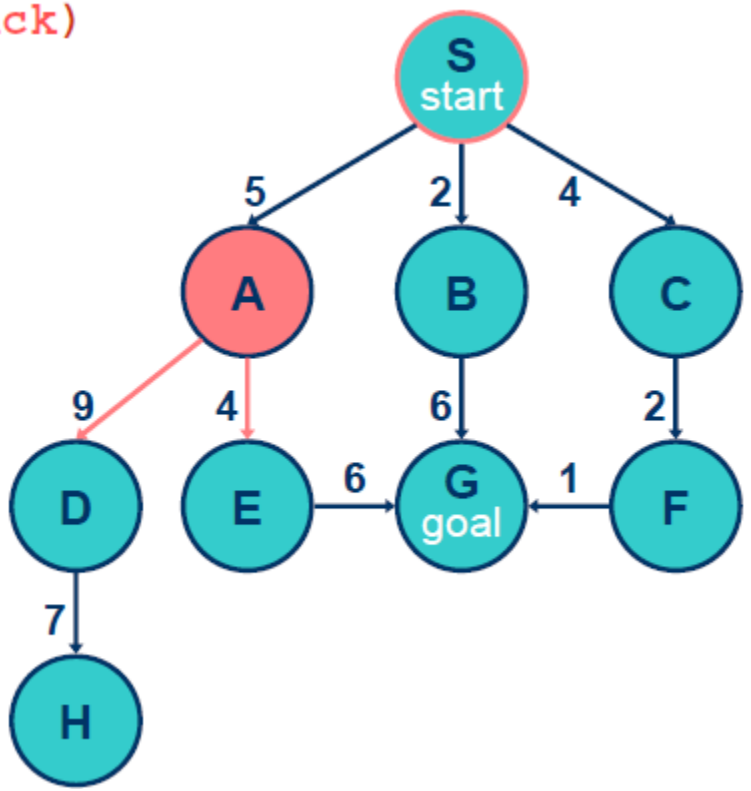
expnd. node	nodes list
	{S}
S not goal	{A,B,C}



`generalSearch(problem, stack)`

of nodes tested: 2, expanded: 2

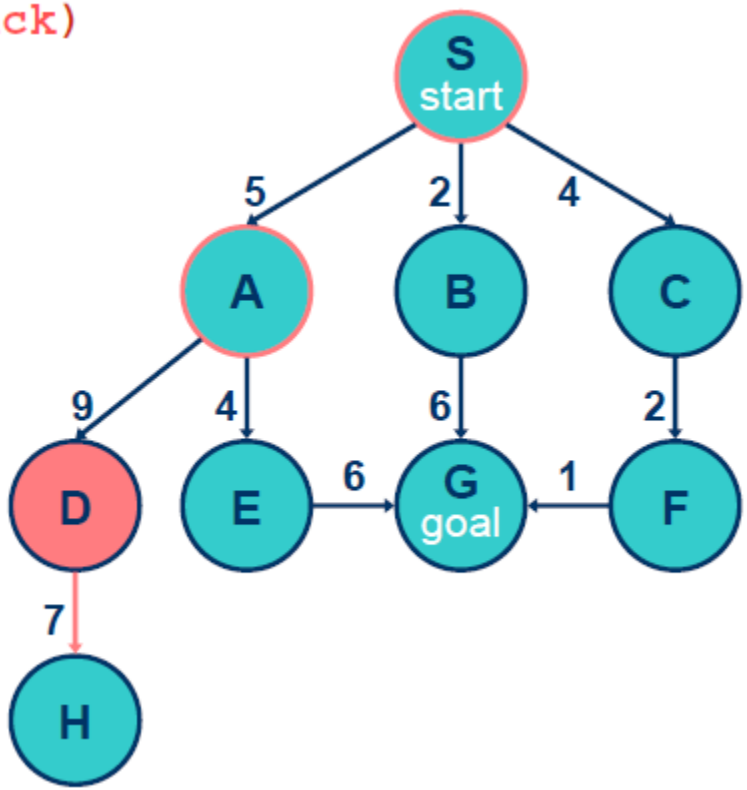
expnd. node	nodes list
	{S}
S	{A,B,C}
A not goal	{D,E,B,C}



`generalSearch(problem, stack)`

of nodes tested: 3, expanded: 3

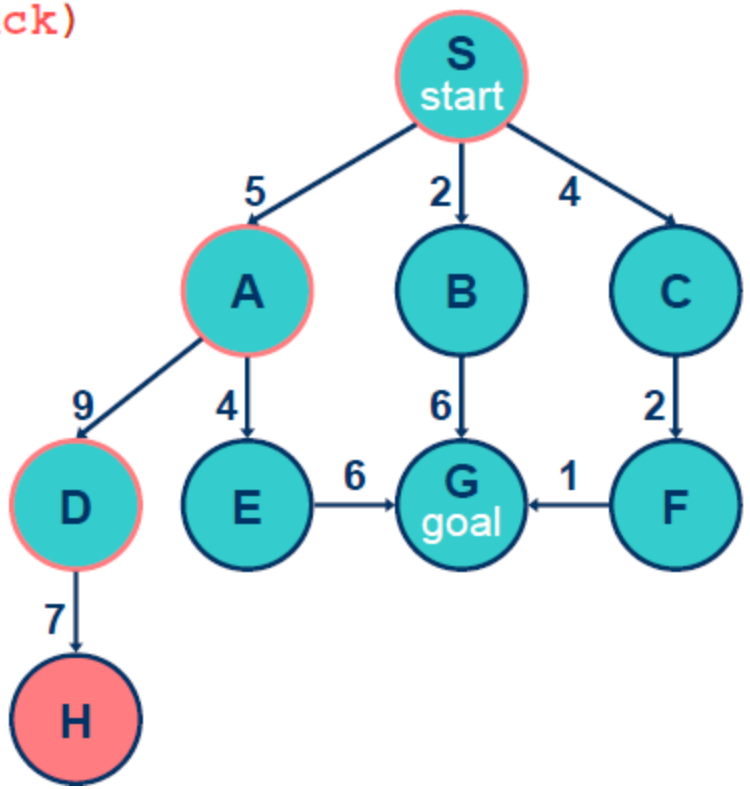
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{D,E,B,C}
D not goal	{H,E,B,C}



`generalSearch(problem, stack)`

of nodes tested: 4, expanded: 4

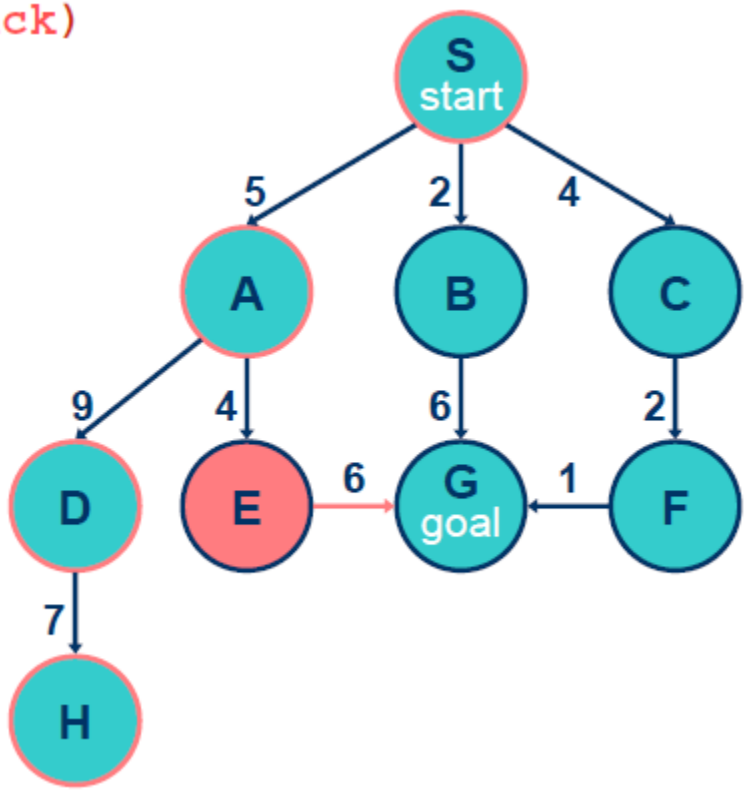
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{D,E,B,C}
D	{H,E,B,C}
H not goal	{E,B,C}



`generalSearch(problem, stack)`

of nodes tested: 5, expanded: 5

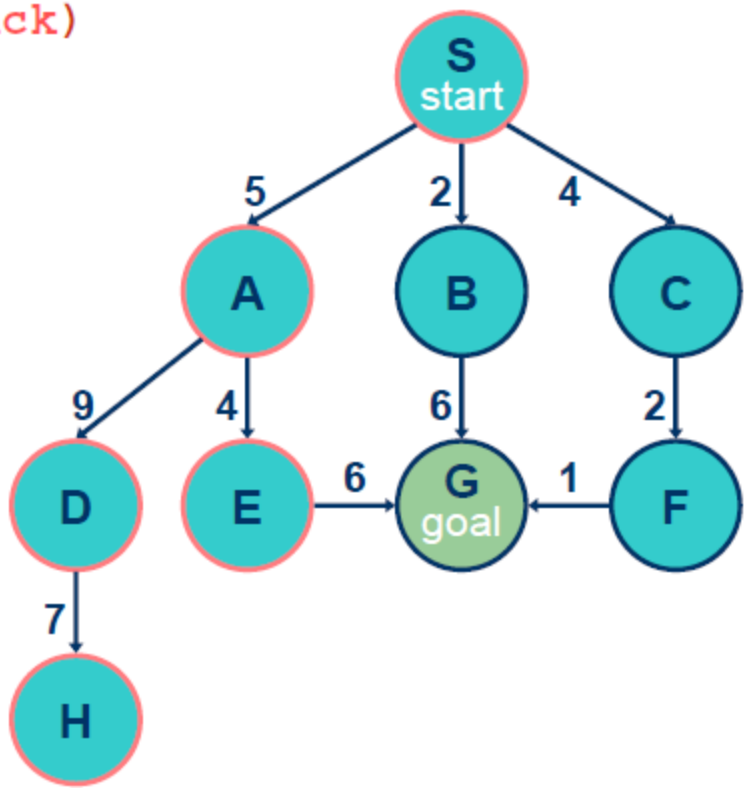
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{D,E,B,C}
D	{H,E,B,C}
H	{E,B,C}
E not goal	{G,B,C}



`generalSearch(problem, stack)`

of nodes tested: 6, expanded: 5

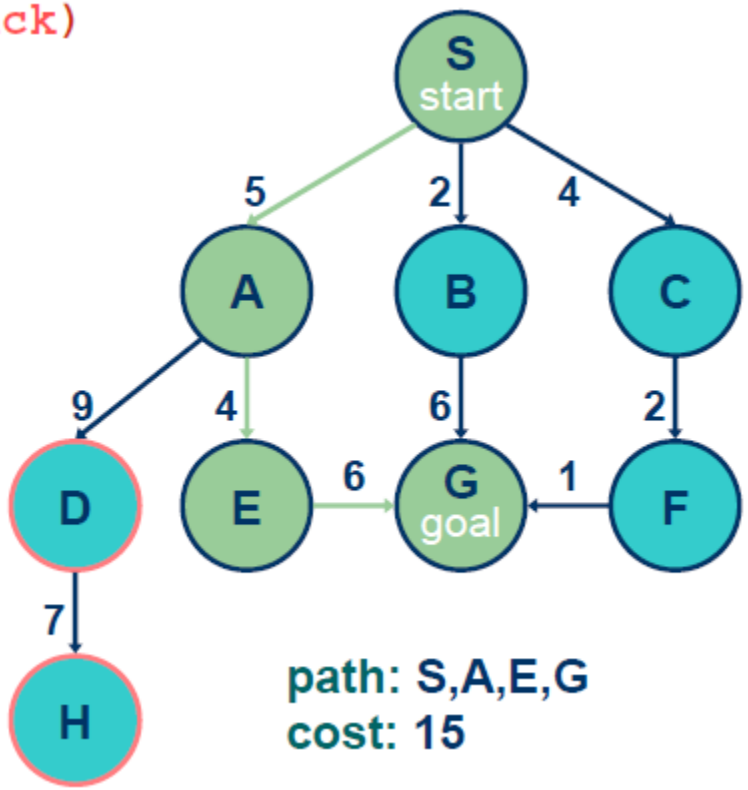
expnd. node	nodes list
	{S}
S	{A,B,C}
A	{D,E,B,C}
D	{H,E,B,C}
H	{E,B,C}
E	{G,B,C}
G goal	{B,C} no expand



`generalSearch(problem, stack)`

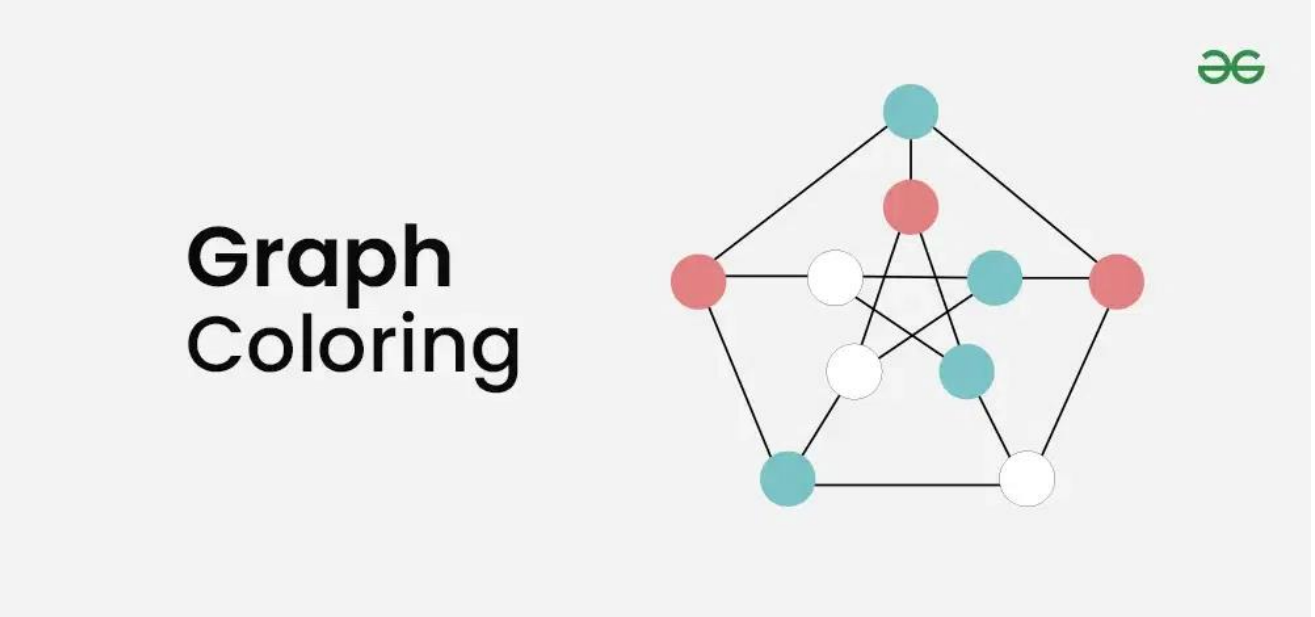
of nodes tested: 6, expanded: 5

expnd. node	nodes list
	{S}
S	{A,B,C}
A	{D,E,B,C}
D	{H,E,B,C}
H	{E,B,C}
E	{G,B,C}
G	{B,C}



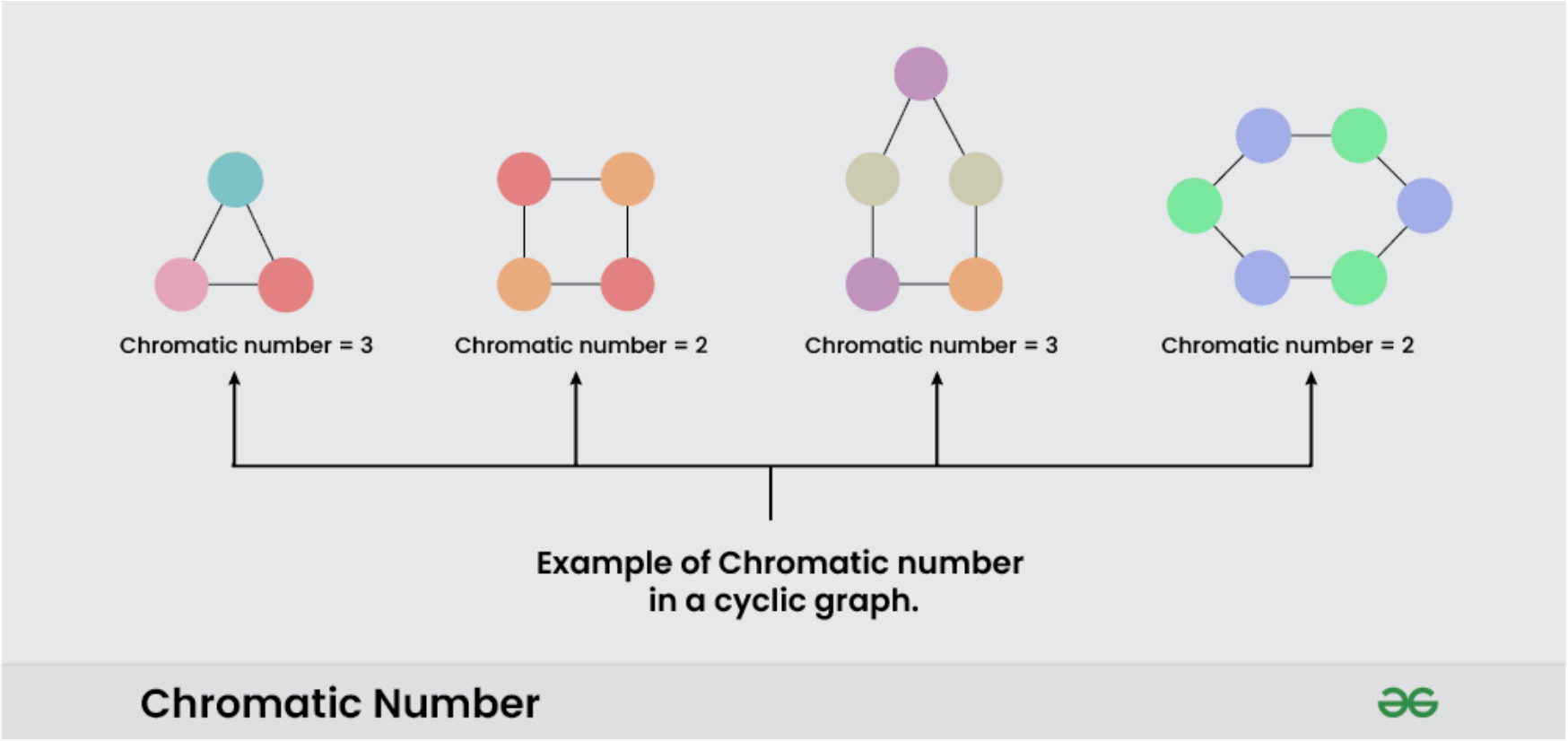
Introduction to Graph Coloring

Graph coloring refers to the problem of coloring vertices of a graph in such a way that no two adjacent vertices have the same color. This is also called the vertex coloring problem. If coloring is done using at most m colors, it is called m-coloring.



Introduction to Graph Coloring

Chromatic Number





Knowledge check



Every binary tree is a binary search tree.



Knowledge check



Every binary search tree is a tree.



Knowledge check



A node in a binary tree must have two children.



Knowledge check



Each node of a binary search tree has a parent.



Knowledge check



In a binary search tree, the info in all of the nodes in the left subtree of a node are less than the info in the node.



Knowledge check



The specially designated node is called the root.



Knowledge check



In a Binary Tree, a node may have a degree greater than 2.



Knowledge check



A tree can represent many-to-many relationships.



Knowledge check



Traversal in a graph is visiting each node at least once.



Knowledge check



Bubble Sort is so named because it bubbles the smallest element to the middle of the array.



Knowledge check



The traversals of a binary search tree differ from the traversals of a binary tree.



Knowledge check



The maximum number of nodes in a binary tree that has N levels is 2^N.



Knowledge check



A binary search starts at the beginning of the collection of items.



Knowledge check



The height of a binary search tree is sensitive to the order in which items are inserted into the tree.



Knowledge check



A Binary Search can only be applied to sorted records.



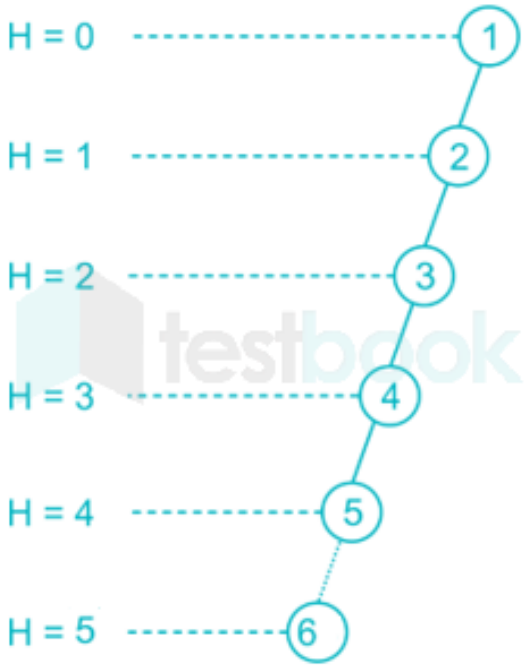
Knowledge check **MCQ**

The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are ……. and ……., respectively



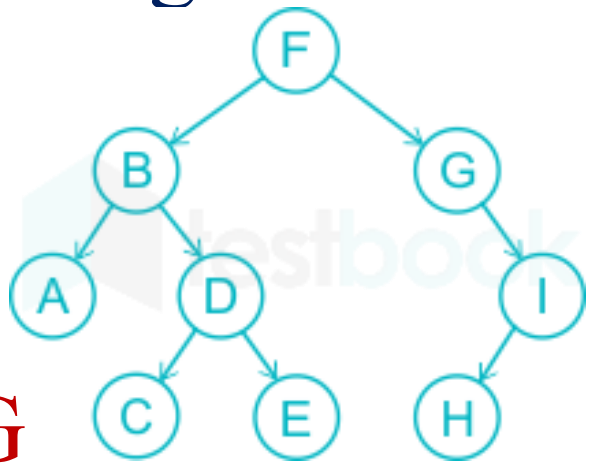
(A) 63 and 6
(B) 64 and 5
(C) 32 and 6
(D) 31 and 5


h = 0
h = 1
h = 2
h = 3
h = 4
h = 5


H = 0
H = 1
H = 2
H = 3
H = 4
H = 5
1
2
3
4
5
6

Knowledge check **MCQ**

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



(A) A, B, C, D, E, F, I, H, G
(B) A, C, D, E, B, F, G, H, I
(C) A, B, C, D, E, F, G, H, I
(D) I, H, G, F, E, D, C, B, A



Knowledge check **MCQ**

Depth First Search graph search algorithm uses …….. data structure for its implementation.



(A) Stack
(B) Queue
(C) Tree
(D) More than one
(E) None



Knowledge check **MCQ**

Breadth First Search graph search algorithm uses ……. data structure for its implementation.

(A) Dequeue



(B) Queue

(C) tree

(D) Stack



Knowledge check

MCQ

The number of swapping needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order, using bubble sort is:

(A) 11
(B) 12
(C) 13
(D) 10



Knowledge check **MCQ**

Binary search works efficiently with a …….. list.

(A) unordered

(B) sorted
(C) linked
(D) circular linked
(E) Queue



Knowledge check

MCQ

Arrange the following items in ascending order using Bubble Sort. What is the intermediate sequence of 37, 54, 21, 85, 68, 12, 9, and 57 after the second pass?

(A) 21, 37, 12, 9, 54, 57, 68, 85
(B) 37, 21, 54, 68, 12, 9, 57, 85
(C) 21, 12, 9, 37, 54, 57, 68, 85
(D) 21, 37, 54, 12, 9, 57, 68, 85

Here are the intermediate steps of the Bubble Sort algorithm for the given list of items: 37, 54, 21, 85, 68, 12, 9, 57

Pass 1

- Compare 37 and 54. No swap.
- Compare 54 and 21. Swap.
- Compare 54 and 85. No swap.
- Compare 85 and 68. Swap.
- Compare 85 and 12. Swap.
- Compare 85 and 9. Swap.
- Compare 85 and 57. Swap.
- **Final sequence: 37, 21, 54, 68, 12, 9, 57, 85**

Pass 2

- Compare 37 and 21. Swap.
- Compare 37 and 54. No swap.
- Compare 54 and 68. No swap.
- Compare 68 and 12. Swap.
- Compare 68 and 9. Swap.
- Compare 68 and 57. Swap.
- Compare 68 and 85. No swap.
- **Final sequence: 21, 37, 54, 12, 9, 57, 68, 85**



Knowledge check

MCQ

Sort the following list using the Radix sort algorithm. [329,457, 839, 436, 720, 355, 657] What is the output of the algorithm after the second pass?

(A) 329, 355, 436, 457, 657, 720, 839
(B) 355, 329, 457, 436, 720, 657, 839

(C) 720, 329, 436, 839, 355, 457, 657
(D) 720, 355, 436, 457, 657, 329, 839

According to the above algorithm -

After Pass 1 - `720, 355, 436, 457, 657, 329, 839`

After Pass 2 - `720, 329, 436, 839, 355, 457, 657`



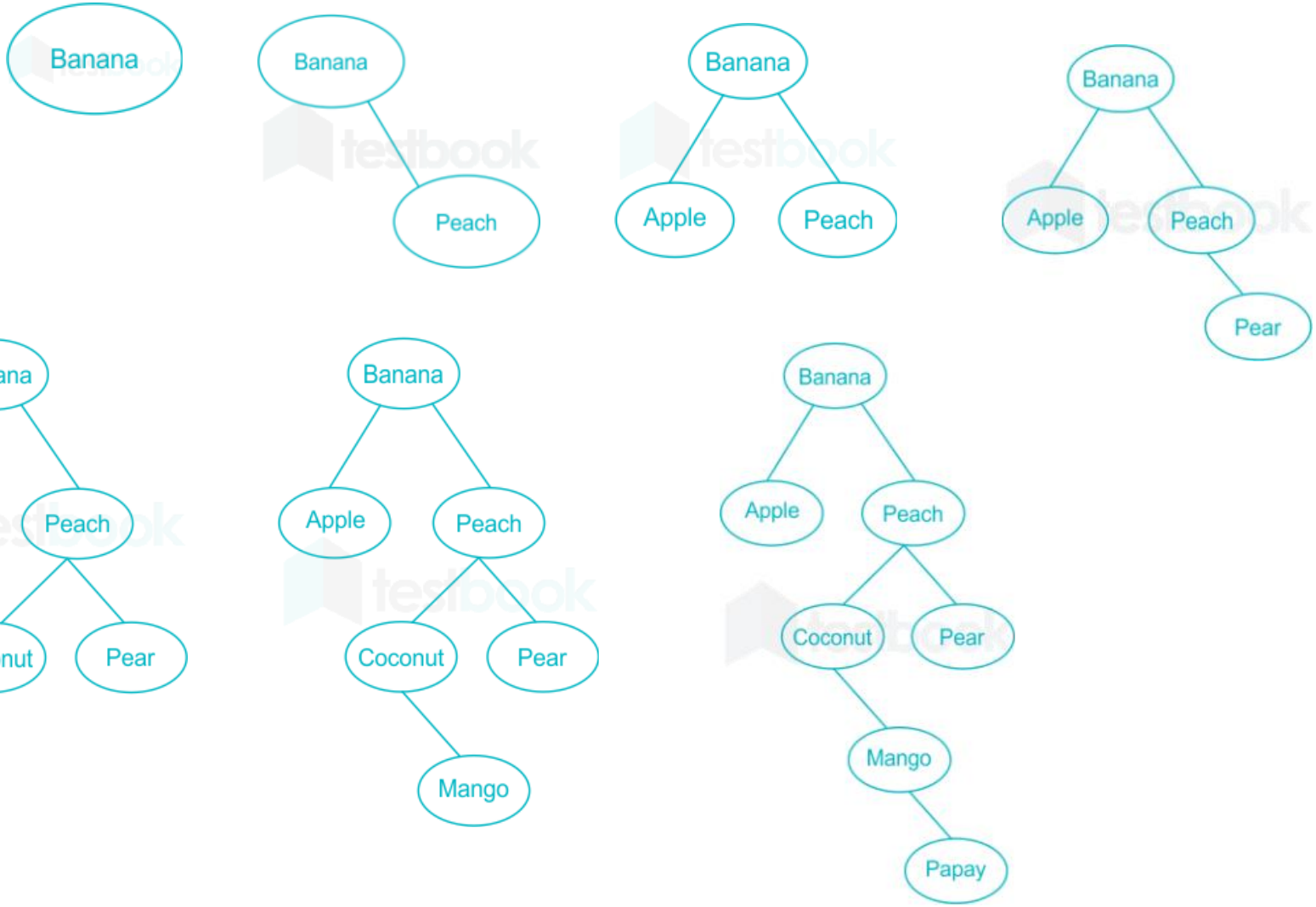


Knowledge check **MCQ**

The height of a binary search tree for the words [banana, peach, apple, pear, coconut, mango and papaya] using alphabetical order is :

(A) 2
(B) 3
(C) 4
(D) 5

[banana, peach, apple, pear, coconut, mango and papaya]





Knowledge check **MCQ**

The maximum height of a binary tree with 'n' nodes is

(A) n

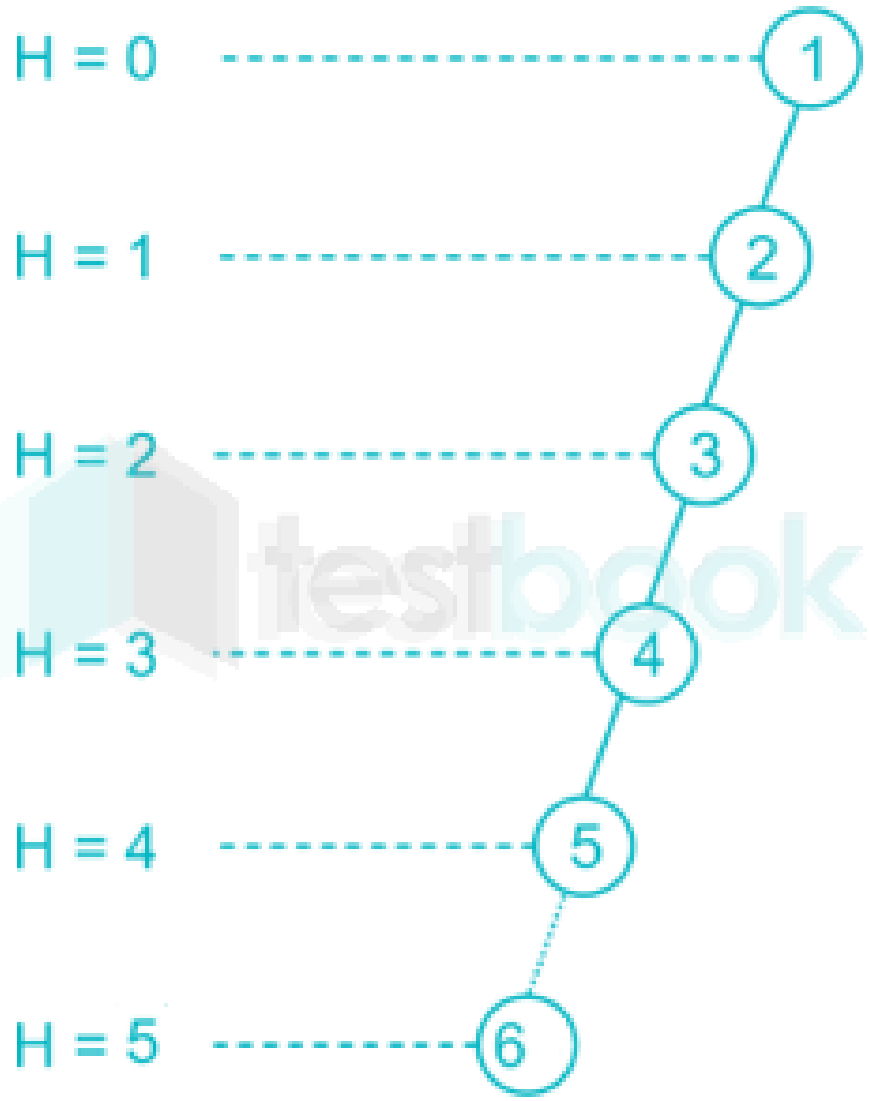


(B) (n-1)

(C) (n + 1)

(D) (n + n)

Tree with maximum height: n = 6



Height = 6 – 1 = 5



Knowledge check **MCQ**

Suppose the keys 3, 4, 45, 21, 92, 12 are inserted into a BST in this order. What is preorder traversal?

(A) 3 4 12 21 45 92



(B) 3 4 45 21 12 92

(C) 12 21 92 45 4 3

(D) 12 92 21 4 45 3



Knowledge check **MCQ**

The Non leaf nodes of a tree are also known as _____ nodes.



(A) Internal
(B) external
(C) outer
(D) root



Knowledge check **MCQ**

A node that does not have any child nodes is called.

(A) Empty
(B) Root
(C) Child
(D) Leaf

End